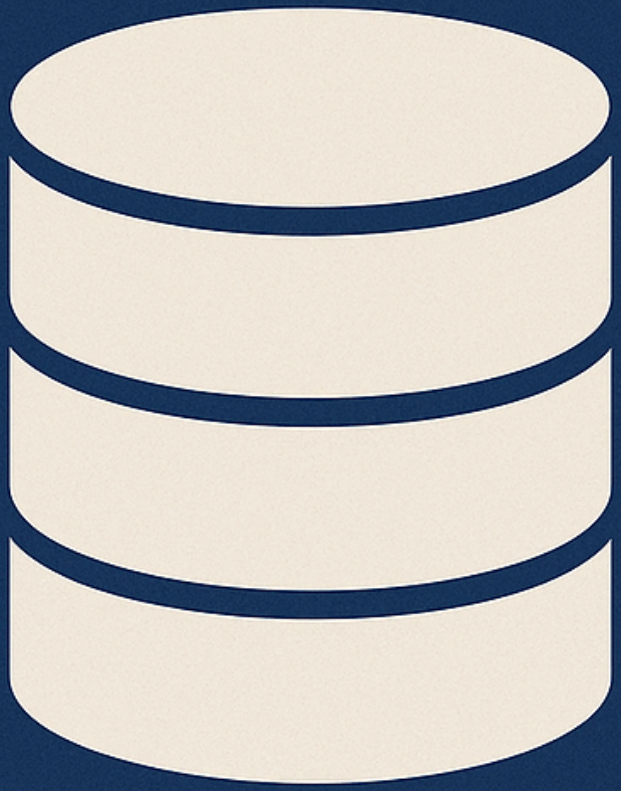
SQL

Class starts
@ 9:10pm





Lecture 7: SQL Joins

#Agenda

1) problem statement
2) Intro to joins
3) Types of Joins → Inner Join / Left Join / Right Join / Full Outer Join

Syntax

Top wear

Select
ColA, ColB, ...

} No of Cols to display

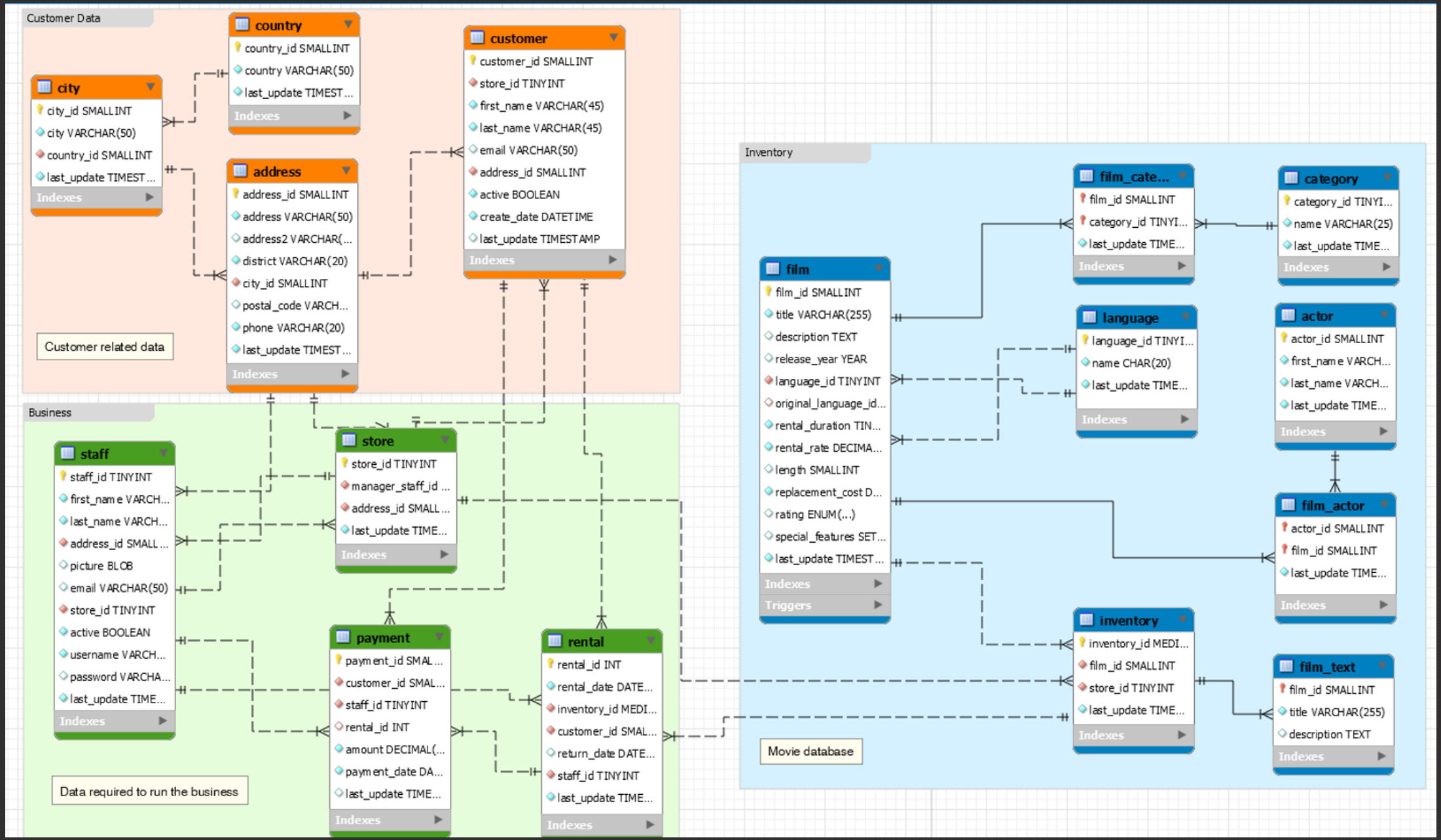
Bottom wear

From
TableA

where
group by
having
limit

order by

} No of rows to be displays


Customer Data
country
country_id SMALLINT
country VARCHAR(50)
last_update TIMEST...
Indexes
customer
customer_id SMALLINT
store_id TINYINT
first_name VARCHAR(45)
last_name VARCHAR(45)
email VARCHAR(50)
address_id SMALLINT
active BOOLEAN
create_date DATETIME
last_update TIMESTAMP
Indexes
city
city_id SMALLINT
city VARCHAR(50)
country_id SMALLINT
last_update TIMEST...
Indexes
address
address_id SMALLINT
address VARCHAR(50)
address2 VARCHAR(...
district VARCHAR(20)
city_id SMALLINT
postal_code VARCH...
phone VARCHAR(20)
last_update TIMEST...
Indexes
Customer related data
Business
staff
staff_id TINYINT
first_name VARCH...
last_name VARCH...
address_id SMALL...
picture BLOB
email VARCHAR(50)
store_id TINYINT
active BOOLEAN
username VARCH...
password VARCHA...
last_update TIME...
Indexes
store
store_id TINYINT
manager_staff_id ...
address_id SMALL...
last_update TIME...
Indexes
payment
payment_id SMAL...
customer_id SMAL...
staff_id TINYINT
rental_id INT
amount DECIMAL(...
payment_date DA...
last_update TIME...
Indexes
rental
rental_id INT
rental_date DATE...
inventory_id MEDI...
customer_id SMAL...
return_date DATE...
staff_id TINYINT
last_update TIME...
Indexes
Data required to run the business
Inventory
film
film_id SMALLINT
title VARCHAR(255)
description TEXT
release_year YEAR
language_id TINYINT
original_language_id...
rental_duration TIN...
rental_rate DECIMA...
length SMALLINT
replacement_cost D...
rating ENUM(...)
special_features SET...
last_update TIMEST...
Indexes
Triggers
film_cate...
film_id SMALLINT
category_id TINYI...
last_update TIME...
Indexes
category
category_id TINYI...
name VARCHAR(25)
last_update TIME...
Indexes
language
language_id TINYI...
name CHAR(20)
last_update TIME...
Indexes
actor
actor_id SMALLINT
first_name VARCH...
last_name VARCH...
last_update TIME...
Indexes
film_actor
actor_id SMALLINT
film_id SMALLINT
last_update TIME...
Indexes
inventory
inventory_id MEDI...
film_id SMALLINT
store_id TINYINT
last_update TIME...
Indexes
film_text
film_id SMALLINT
title VARCHAR(255)
description TEXT
Indexes
Movie database

Joins → Connecting two or more tables

(SQL Join)

↓

Connection

↓

PK & FK

(Keys)

(parent)
Dad (relation)

↙ ↘

C_1 C_2

table table

Neighbour Dad

↓

C_3

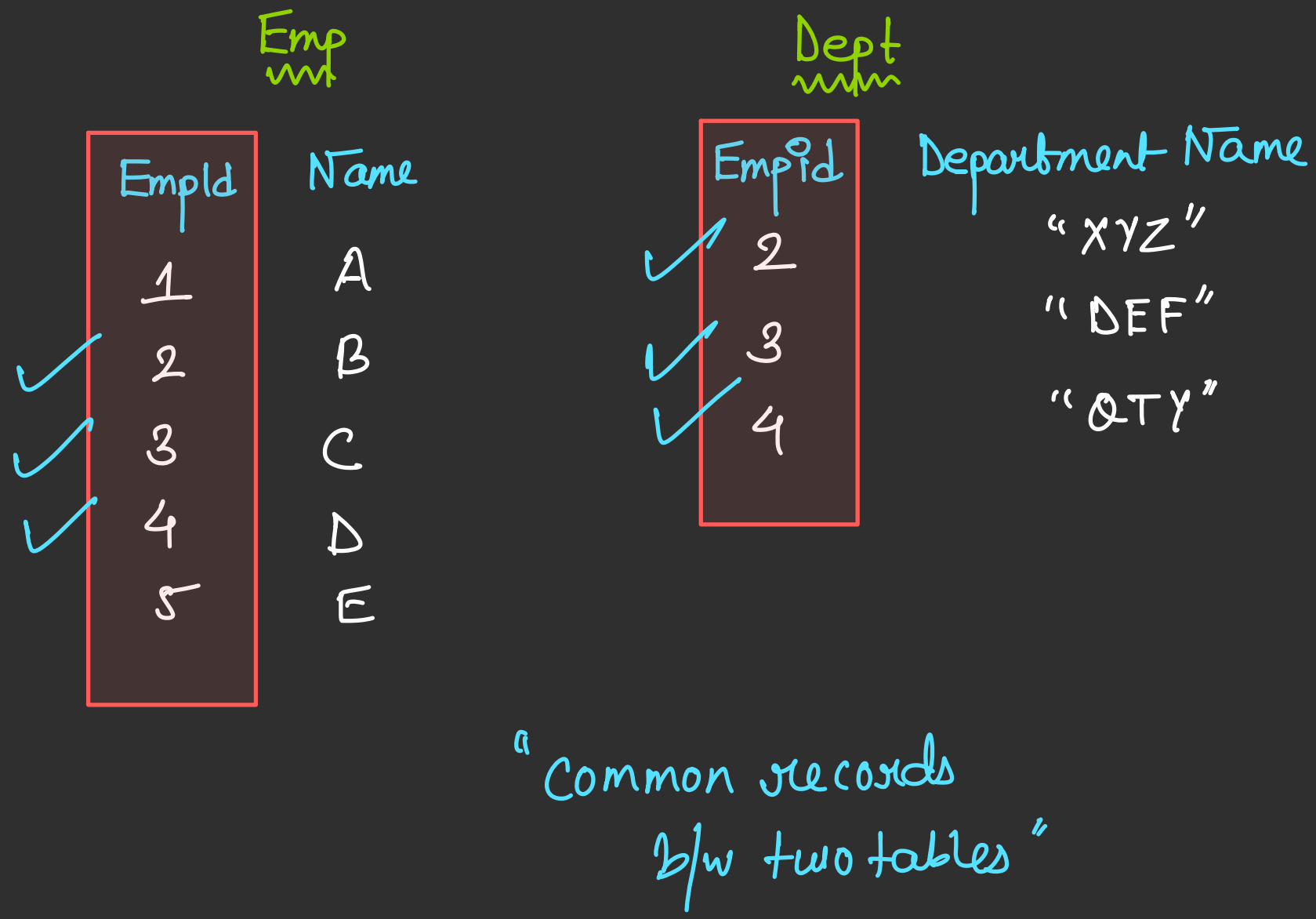
Table 1



Table 2



JOINS


Emp
EmpId
Name
1
A
2
B
3
C
4
D
5
E
Dept
EmpId
Department Name
2
"XYZ"
3
"DEF"
4
"QTY"
"Common records
b/w two tables"

Emp

EmpId	Name
1	A
2	B
3	C
4	D
5	E

Dept

EmpId	Department Name
2	"XYZ"
3	"DEF"
4	"QTY"

W/a "common records b/w two tables"

Query o/p

EmpId	Name	DeptNa
2	B	XYZ
3	C	DEF
4	D	QTY

Syntax

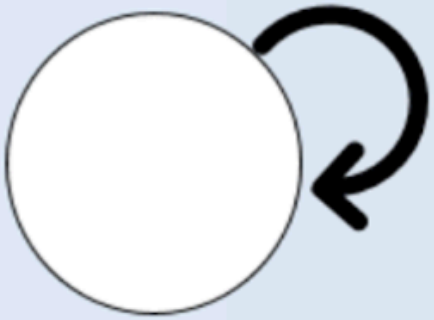
```
Select
  *
From
emp e
INNER JOIN
dept d
ON e.empid = d.empid
```

Query o/p

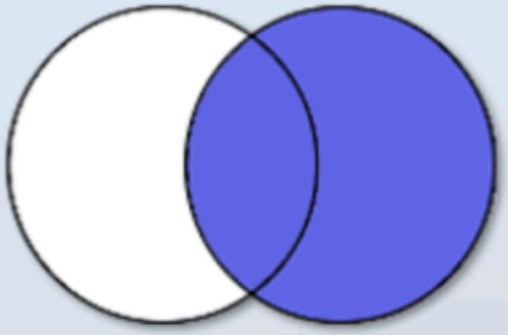
EmpId	Name	DeptNa
2	B	XYZ
3	C	DEF
4	D	QTY

Joins in MySQL

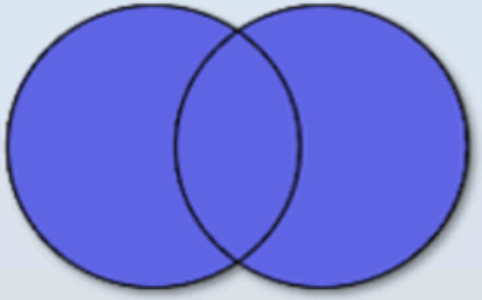
Self Join



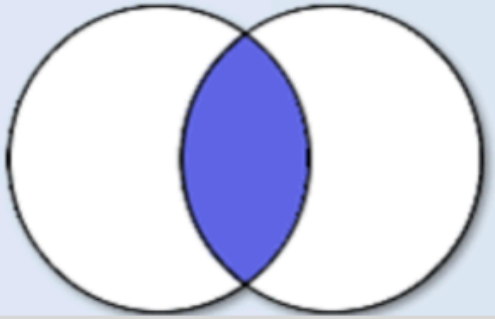
Right Join



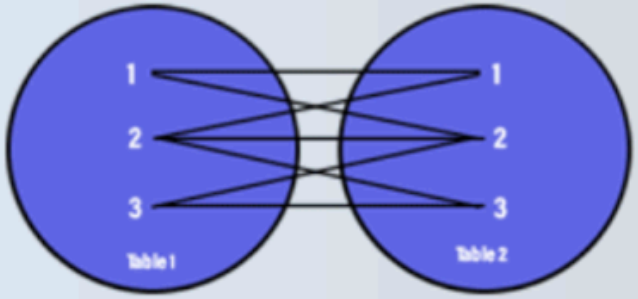
Full Join



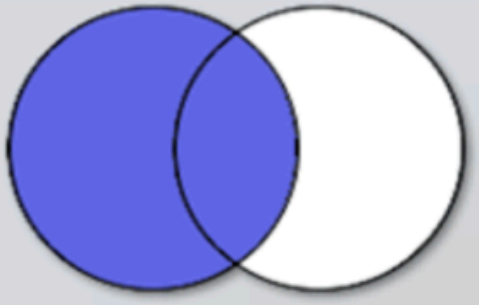
Inner Join



Cross Join



Left Join



How does Joins works?

Table A

ColA	ColB	ColC
		1
		3
		2

Table B

ColA	ColB	ColC
		1
		2
		8

perform Join → increasese/decreases no of rows

Table A

ColA	ColB	ColC
Name	Add	1
		3
		2

Table B

ColA	ColB	ColC
MNam	Hth	1
		2
		8



Table A

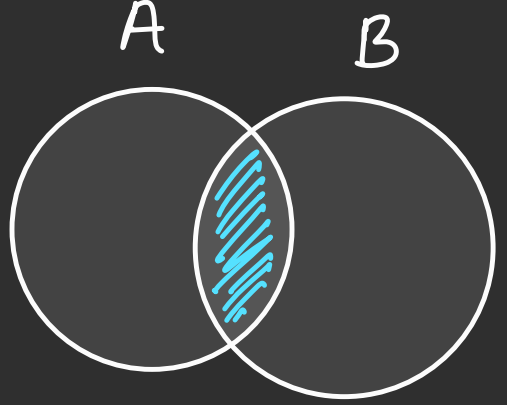
ColA	ColB	ColC
Name	Add	1
		3
		2

Table B

ColA	ColB	ColC
MNbm	HH	1
		2
		2

Inner join → 3 rows

↳ All common records b/w two tables.



Syntax

```
Select
[Col_Name]
from
[Table_Name A]
[Join Type]
[Table_Name B]
ON [Table_Nam A].[Col] = [Table_Name B].[Col]
```

Stores Table (Left Table)

store_id	store_name
1	Store A
2	Store B
2	Store B2
3	Store C
4	Store D

Customers Table (Right Table)

customer_id	customer_name	store_id	active
1	Alice	1	1
2	Bob	2	0
3	Charlie	NULL	1
4	Dave	3	1
5	Eve	3	1
6	Frank	2	1

Result

customer_id	customer_name	store_id	active	store_id	store_name
1	Alice	1	1	1	Store A
2	Bob	2	0	2	Store B
2	Bob	2	0	2	Store B2
4	Dave	3	1	3	Store C
5	Eve	3	1	3	Store C
6	Frank	2	1	2	Store B
6	Frank	2	1	2	Store B2

(Inner join)

Table A

Name	ID
	1
	Null
	2

Inner join

Table B

Store	ID	
	1	
	Null	✗
	Null	✗

No of rows → 1 mn

Stores Table (Left Table)

store_id	store_name
1	Store A
2	Store B
2	Store B2
3	Store C
4	Store D

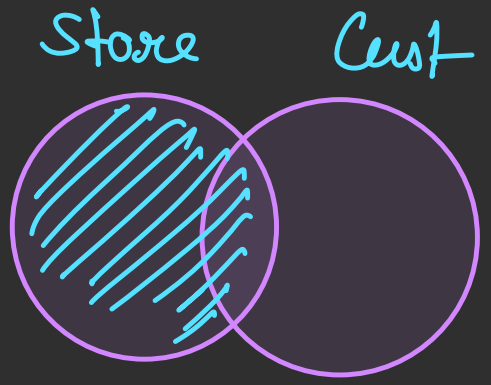
Customers Table (Right)

customer_id	customer_name	store_id	active
1	Alice	1	1
2	Bob	2	0
3	Charlie	NULL	1
4	Dave	3	1
5	Eve	3	1
6	Frank	2	1

Result

customer_id	customer_name	store_id	active	store_id	store_name
1	Alice	1	1	1	Store A
2	Bob	2	0	2	Store B
6	Frank	2	1	2	Store B
2	Bob	2	0	2	Store B2
6	Frank	2	1	2	Store B2
4	Dave	3	1	3	Store C
5	Eve	3	1	3	Store C
NULL	NULL	NULL	NULL	4	Store D

(Left Join)



Stores Table (left Table)

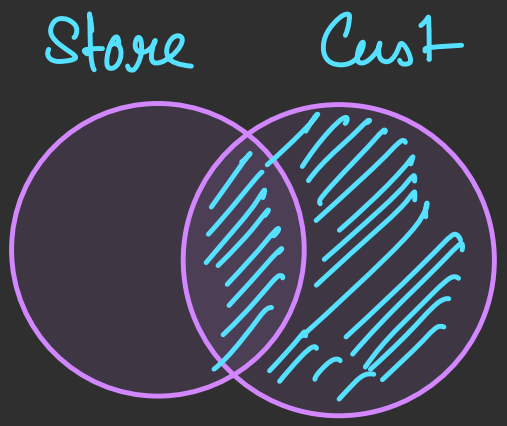
store_id	store_name
1	Store A
2	Store B
2	Store B2
3	Store C
4	Store D

Customers Table (Right Table)

customer_id	customer_name	store_id	active
1	Alice	1	1
2	Bob	2	0
3	Charlie	NULL	1
4	Dave	3	1
5	Eve	3	1
6	Frank	2	1

Result

customer_id	customer_name	store_id	active	store_id	store_name
1	Alice	1	1	1	Store A
2	Bob	2	0	2	Store B
2	Bob	2	0	2	Store B2
3	Charlie	NULL	1	NULL	NULL
4	Dave	3	1	3	Store C
5	Eve	3	1	3	Store C
6	Frank	2	1	2	Store B
6	Frank	2	1	2	Store B2



Select

↗

from

Table A → Cust

Left Join

Table B → Order

ON - - - = - - - -

Select

↗

from

Table A

Right Join

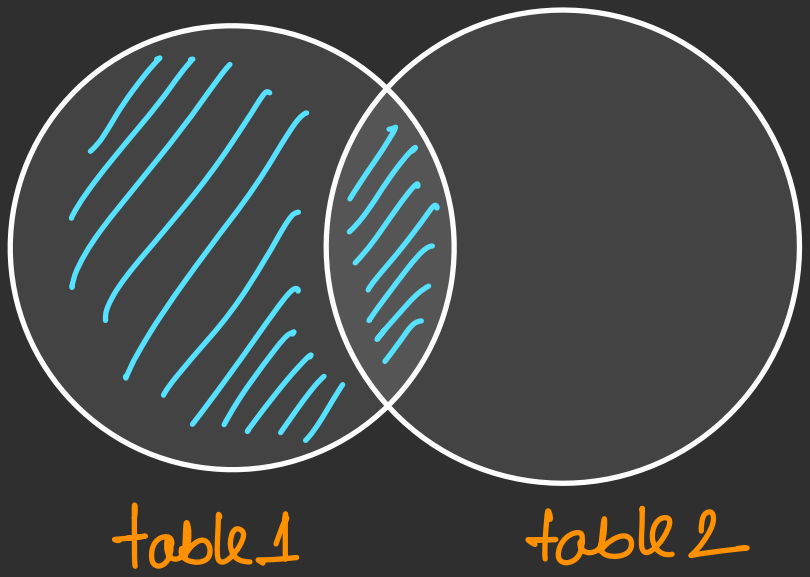
Table B

ON - - - = - - - -

Result

customer_id	customer_name	store_id	active	store_id	store_name
1	Alice	1	1	1	Store A
2	Bob	2	0	2	Store B
2	Bob	2	0	2	Store B2
3	Charlie	NULL	1	NULL	NULL
4	Dave	3	1	3	Store C
5	Eve	3	1	3	Store C
6	Frank	2	1	2	Store B
6	Frank	2	1	2	Store B2
NULL	NULL	NULL	NULL	4	Store D

Left Join




```
From
    table_1 as t1
    LEFT JOIN
    table_2 as t2
    ON t1.col = t2.col
```

left → table_1 as t1

right → table_2 as t2

Homework

1) Inner Join

2) Left Join / Right Join